Analysis of how welfare check logistics shifted post-CAHOOTS

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All cleaning, analysis, and project logistics are available through this GitHub link.

1 Background

When in service in Eugene, Oregon, a large portion of CAHOOTS calls were welfare checks. Since CAHOOTS stopped services in Eugene, how have welfare checks been handled (by EPD? MCSLC?)? Has welfare check call volume changed? Has there been a change in the number of arrests during these welfare check calls? Springfield will be used as a control.

Since CAHOOTS is no longer in operation, most if not all welfare check calls are dispatched directly to EPD and MCSLC. The community had trust in CAHOOTS, which is why they called them for welfare checks in the first place. To date, no study has examined how the removal of a co-responder program affects welfare check call volume and outcomes. This analysis is the first to address this gap. This leads to the research questions:

1. How has the overall volume of welfare check calls changed since April, 2025?
2. Among these calls, how has the number of arrest outcomes changed since April, 2025?

These findings could inform people why the presence of CAHOOTS was crucial and could help law enforcement better approach welfare checks.

The significance of these questions lies in understanding how the defunding of CAHOOTS affected welfare check outcomes in Eugene. By using Springfield as a control, this study will provide evidence for why CAHOOTS is a critical resource for Eugene, informing how CAHOOTS/WVCC can make the case for restored funding and improved collaboration with EPD.

2 Data

Before beginning to describe the data used in this project, all of the data cleaning and reading process was done in a file called `clean_data.ipynb` which is located in my GitHub through clicking on this link In order to address the research questions, I used 3 publicly available datasets:

1. Computer Assisted Dispatch (CAD) of Eugene dataset
2. Springfield 911 call dataset
3. Springfield 911 call outcome dataset

In order to get the initial datasets read and correct, I first read them in using the following snippet (Figure 1).

```python
# read all data in the data folder; will take ~4 minutes to run
path = "data/"

xl = pd.ExcelFile(os.path.join(path, "2015-2025 SPD Calls with Close Codes.xlsx"))

# for the SPD Outcomes data, which includes an unnessary "KEY" sheet that we will ignore
sheets = [sheet for sheet in xl.sheet_names if sheet != "KEY"]

print("Reading CSV")
print('2015...')
eug_cad2015 = pd.read_csv(os.path.join(path, "EugeneCAD2015noloc.csv"))
print('2016...')
eug_cad2016 = pd.read_csv(os.path.join(path, "EugeneCAD2016noloc.csv"))
print('2017...')
eug_cad2017 = pd.read_csv(os.path.join(path, "EugeneCAD2017noloc.csv"))
print('2018...')
eug_cad2018 = pd.read_csv(os.path.join(path, "EugeneCAD2018noloc.csv"))
print('2019...')
eug_cad2019 = pd.read_csv(os.path.join(path, "EugeneCAD2019noloc.csv"))
print('2020...')
eug_cad2020 = pd.read_csv(os.path.join(path, "EugeneCAD2020noloc.csv"))
print('2021...')
eug_cad2021 = pd.read_csv(os.path.join(path, "EugeneCAD2021noloc.csv"), low_memory=False)
print('2022...')
eug_cad2022 = pd.read_csv(os.path.join(path, "EugeneCAD2022noloc.csv"))
print('2023...')
eug_cad2023 = pd.read_csv(os.path.join(path, "EugeneCAD2023noloc.csv"))
print('2024...')
eug_cad2024 = pd.read_csv(os.path.join(path, "EugeneCAD2024noloc.csv"))
print('2025...')
eug_cad2025 = pd.read_csv(os.path.join(path, "EugeneCAD2025noloc.csv"), low_memory=False)

print("Reading Excel")
print('SPD Calls for Service...')
spd_calls = pd.concat(pd.read_excel(os.path.join(path, '2015-2025 SPD Calls for Service.xlsx'), sheet_name=None), ignore_index=True)
print('SPD Responding Units...')
spd_units = pd.concat(pd.read_excel(os.path.join(path, '2015-2025 SPD Responding Units.xlsx'), sheet_name=None), ignore_index=True)
print("SPD Outcomes...")
spd_outcomes = pd.concat(pd.read_excel(os.path.join(path, '2015-2025 SPD Calls with Close Codes.xlsx'), sheet_name=sheets), ignore_index=True)
print("Done")
```

Figure 1: Reading in datasets

After, I put all the Eugene datasets together using the following snippet (Figure 2).

```python
dfs = [
    eug_cad2015, eug_cad2016, eug_cad2017, eug_cad2018,
    eug_cad2019, eug_cad2020, eug_cad2021, eug_cad2022,
    eug_cad2023, eug_cad2024, eug_cad2025
]

base_cols = set(dfs[0].columns)

for i, df in enumerate(dfs):
    cols = set(df.columns)
    missing = base_cols - cols
    extra = cols - base_cols

    if missing or extra:
        print(f"DataFrame {2015 + i}:")
        if missing:
            print("  Missing:", missing)
        if extra:
            print("  Extra:", extra)

# drop that extra column
eug_cad2025 = eug_cad2025.drop(columns={"month"})
```

Figure 2: Organizing Eugene dataset

Below is a description of the datasets.

2.1 CAD of Eugene Dataset

`OUTPUT : cleaned_eug.csv`

Each row is a 911 call assigned to either Eugene Police Department (EPD) or CAHOOTS. Key variables include call nature (`Nature`), time data, call outcome, dispatched unit, and the agency initially called. The dataset contains ~1.5 million individual call records, making it well-suited for data visualization. Time-based variables are easily converted to datetime objects. Some time-related columns contain missing values; however, the only time column used in this analysis is `calltime`, which is complete so the missing time data will not impact the analysis.

2.2 Springfield 911 Call Dataset

`OUTPUT : cleaned_spd.csv`

Each row is a call made to Springfield Police Department (SPD). Key variables include call nature, responding agency and unit, time data, call priority, and call creation method. Call nature is more clearly defined here than in the Eugene dataset, with fewer category options. Time-based variables are easily converted to datetime objects. The `Call Creation Mechanism` column contains ambiguous values (PHONE, SELF, W911, N911) and its meaning is unclear; however, this column is not used in the analysis. The columns used: `Final Call Type`, `Call Creation Time`, and `Primary Responding Unit` are clearly defined and complete.

2.3 Springfield 911 Call Outcome Dataset

`Used to produce cleaned_spd.csv`

Each row in this dataset represents a specific closing code for a call in the dataset above. It contains a column that has an ID corresponding to a specific call, and a closing code that says how that specific call ended. The true purpose of this dataset is simply to merge it with the dataset above in order to get information of how the calls ended. In doing so, we can address the second research question.

2.4 How was this data prepared for analysis?

All cleaning necessary for these datasets were done in the file called `clean_data.ipynb`, which is available on Github using the link at the beginning of the report.

The packages used for the cleaning are:

1. Pandas - Dataset manipulation

The cleaning began with the CAD Dataset of Eugene. Here were the steps taken:

1. Subset to the columns needed to address the research question (by column name: `calltime`, `nature`, `closed_as`, `primeunit`)
2. Convert the `calltime` column to datetime for easy time manipulation.
3. Further subset the data so that the nature column only includes entries that include `"CHECK WELFARE"`.

(a) Note that this step does result in ~10% of the `primeunit` column to include missing values or NAs. This is completely fine because we do not require information from that column in order to address the research questions.

The resulting data frame is shown in Figure 3.

calltime	nature	closed_as	primeunit
2015-01-01 00:01:05	CHECK WELFARE	ASSISTED	_3J79
2015-01-01 00:38:37	CHECK WELFARE	ASSISTED	_3J79
2015-01-01 01:28:40	CHECK WELFARE	UNABLE TO LOCATE	_5B44
2015-01-01 01:41:40	CHECK WELFARE	ASSISTED	_3J79
2015-01-01 15:09:21	CHECK WELFARE	ASSISTED	_3J79

Figure 3: Cleaned CAD Dataset

For the Springfield 911 Call Dataset, the following steps were followed:

1. Merge the Springfield Call Outcome dataset with this one (there is no cleaning needing to be done on that dataset)

2. Subset to the columns needed to address the research question (by column name: `Final Call Type`, `Call Creation Time`, `Primary Responding Unit`, and `Close Code`)

3. Subset the dataset further to only include welfare check calls (all entries in `Final Call Type` should be `CHECK WELFARE`)

4. Rename columns within dataset to match that of the CAD dataset of Eugene for consistency (optional)

5. Turn the `Call Creation Time` column into datetime for time manipulation

6. Remove any errors in the `Primary Responding Unit` and `Close Code` columns (such as a string of random numbers being present)

The resulting dataframe is in Figure 4.

nature	calltime	primeunit	closed_as
CHECK WELFARE	2015-01-04 19:27:33	3S21	WELC
CHECK WELFARE	2015-01-01 04:07:43	1S22	ASST
CHECK WELFARE	2015-01-01 12:32:22	2S11	UTL
CHECK WELFARE	2015-01-02 09:12:29	2S11	UNFD
CHECK WELFARE	2015-01-02 16:06:16	3S11	REPT

Figure 4: Cleaned SPD Dataset

Note that this dataset shows relatively the same thing as the Eugene CAD dataset, though in a different order with different entries.
For the Springfield 911 Call Outcome data, no cleaning was needed and no processing was required for this dataset. The only use was to merge it with the SPD dataset mentioned above.

The resulting dataframe is in Figure 5.

Incident Number	Close Code
15150678	ABAN
15201625	ABAN
15276199	ABAN
15008945	ACC
15030803	ACC

Figure 5: SPD Outcome Dataset

3 Methods

In terms of how I used my data to address my research questions, I used difference in differences. The reason I used this method is because we have two very comparable groups:

1. Eugene - No CAHOOTS [Treatment Group]

2. Springfield - CAHOOTS [Control Group]

Through using this method, I can learn the impact CAHOOTS being defunded on:

1. Volume of welfare check calls

2. Volume of arrest outcomes

which is exactly what my research questions were surrounding.
There will be two unique difference in differences plots; one for each research question. Here is how both of them will be created.

3.1 Welfare Check Call Volume Analysis

`NOTE : All processes discussed in this section are located in volume_analysis.ipynb on GitHub through clicking on this link`

1. Packages used: `numpy`, `pandas`, `matplotlib`, `scipy`

2. Input: Cleaned Eugene CAD and Springfield datasets

3. Output: `diff_in_diff.png`, p-value from a t-test

4. Steps

   (a) Group call data by week to measure weekly welfare check call volume.

   (b) Conduct difference-in-differences (DiD) analysis with the help of matplotlib and pandas. The x-axis is the date and the y-axis is the weekly welfare check call volume. A red, vertical, dashed line marks the date of CAHOOTS being defunded (April 7th, 2025), with the post-defunding period representing the change in weekly welfare check call volume. Eugene will be in a brownish color, Springfield in a grayish color, and a counterfactual line for the analysis in a blue-ish color. The counterfactual was created using a regression that took the previous 30 weeks worth of data before CAHOOTS was defunded.

(c) Conduct a one-tailed independent sampled t-test to ensure that the results of the difference-in-differences analysis are statistically significant. The null hypothesis is the weekly volume of welfare check calls before and after defunding are the relatively the same; the alternative is that post-defunding weekly call volume is smaller. A t-test is appropriate in this scenario because a comparison between averages is being made across two independent time periods. Below is the method used to find the p-value for this t-test (Figure 6).

```python
before_treatment = merged[merged["calltime"] < treatment_date]["count_eug"]
after_treatment = merged[merged["calltime"] > treatment_date]["count_eug"]

# Perform test
t_val, p = sci.ttest_ind(before_treatment, after_treatment, equal_var=False, nan_policy='omit', alternative="greater")

print(f"P-value: {p}")

```

Figure 6: Method to find p-value for volume analysis

3.2 Arrest Outcome Volume Analysis

`NOTE : All processes discussed in this section are located in outcome_analysis.ipynb on GitHub through clicking on this link`

1. Packages used: `pandas`, `matplotlib`, `numpy`, `scipy`

2. Input: Cleaned Eugene CAD and Springfield datasets

3. Output: `diff_in_diff_outcome.png`, p-value from a t-test

4. Steps

   (a) Group call outcome data by month to measure monthly arrest outcomes coming from welfare checks.

   (b) Conduct difference-in-difference analysis with the help of matplotlib and pandas. The x-axis is the date and the y-axis is the monthly average of arrest outcomes. A red, vertical, dashed line marks the date of CAHOOTS being defunded, with the post defunding period representing the change in monthly average arrest outcome volume. Eugene will again be the in a brown-ish color. Springfield in a gray-ish color, and a counterfactual line for the analysis in a blue-ish color. The counterfactual was created through using the previous 5 months of data before CAHOOTS was defunded.

(c) Conduct a one-tailed independent sampled t-test to ensure that the results of the difference-in-differences analysis are statistically significant. The null hypothesis here is that the monthly average volume of arrest outcomes is relatively the same; the alternative is that after defunding, monthly arrest outcomes are larger. A t-test is once again appropriate in this scenario because a comparison between averages is being made across two independent time periods. Below is the method used to find the p-value for this t-test (Figure 7).

```python
before_treatment = merged[merged["calltime"] < treatment_date]['count_eug']
after_treatment = merged[merged["calltime"] > treatment_date]['count_eug']
# Perform test
t_val, p = sci.ttest_ind(before_treatment, after_treatment, equal_var=False, nan_policy='omit', alternative='less')

print(f"P-value: {p}")
```

Figure 7: Method to find p-value for arrest volume analysis

4 Results

Onto the results. From both the volume and outcome analyses, the DiD analysis and t-test were both successful. Here is what I found.

4.1 Volume

The plot below (Figure 8) shows a line plot showcasing the DiD analysis for the volume analysis. We see from this plot that both Eugene and Springfield (with Eugene in the brown-ish color and Springfield in the gray-ish color) show similar trends across time up until the point where CAHOOTS was defunded and stopped its operations, depicted by the red-dashed line. After this point, we see that Eugene's line starts to dip while Springfield's stays relatively at the same level. If we take a closer look at the counterfactual line (in the blue-ish color) behind Eugene, we see that we expect Eugene to have a larger weekly volume of welfare check calls, but in reality, we are seeing a sharp decrease in the volume of welfare checks.

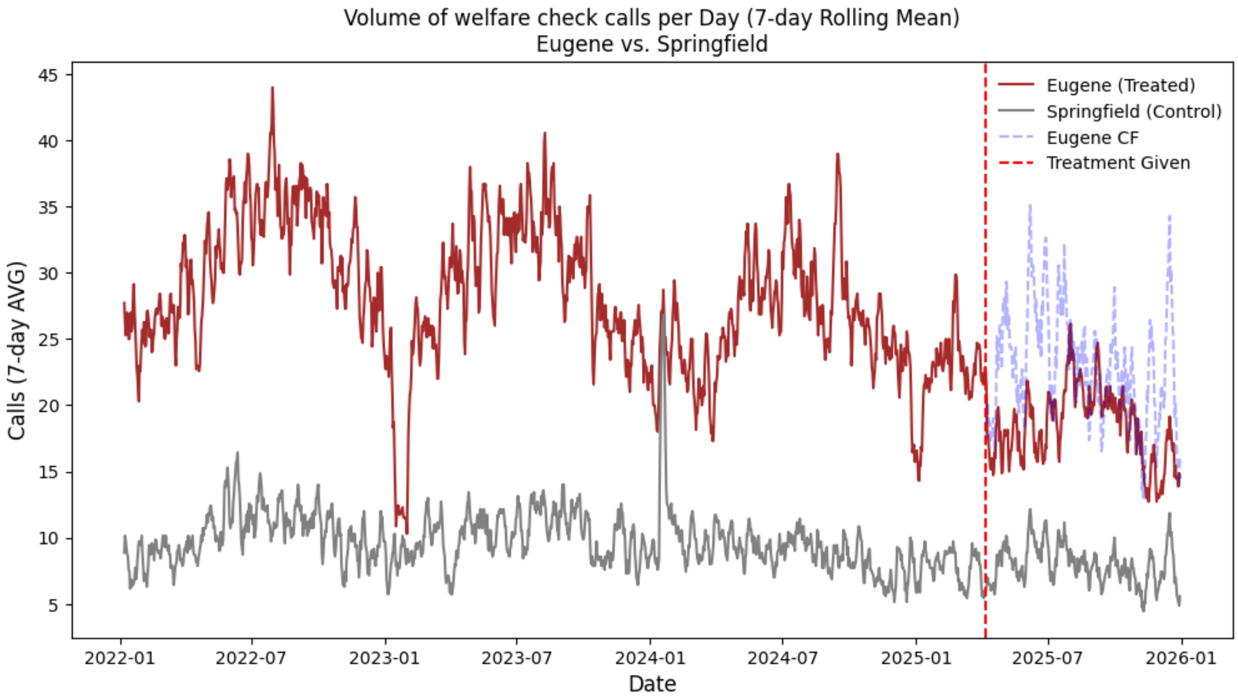

Figure 8: DiD Lineplot #1: Volume

From the t-test conducted, the null hypothesis (as said before) was that the weekly volume of welfare check calls before and after defunding are the relatively the same. After conducting the t-test with a p-value ¡ 0.001 (essentially 0), we can reject the null hypothesis and accept the alternative hypothesis that the post-defunding weekly volume of weekly call volume is smaller, therefore saying that the drop in volume is statistically significant.

4.2 Outcome

The plot below (Figure 9) shows a line plot showcasing the DiD analysis done for the outcome analysis. We see from this plot that it is very different from the previous one used in the volume result section. From this plot, we see that Eugene has an oscillating trend in arrest outcomes from welfare check calls while Springfield's stays relatively the same all throughout. This is interesting, but it's important to remember that Springfield does not have the same volume of calls that Eugene does, which means their numbers will be low almost all the time. However, from what is seen on this plot, Springfield seems to have some "humps" in terms of arrest outcomes throughout its time. After the red-line is met (when CAHOOTS was defunded), we see Eugene see a sharp rise in arrest outcomes, but them begins to level off while Springfield's line stays relatively the same. Taking a look at the counterfactual, the expectation is that there should be far less arrest outcomes, but in reality there are far more arrest outcomes in Eugene.

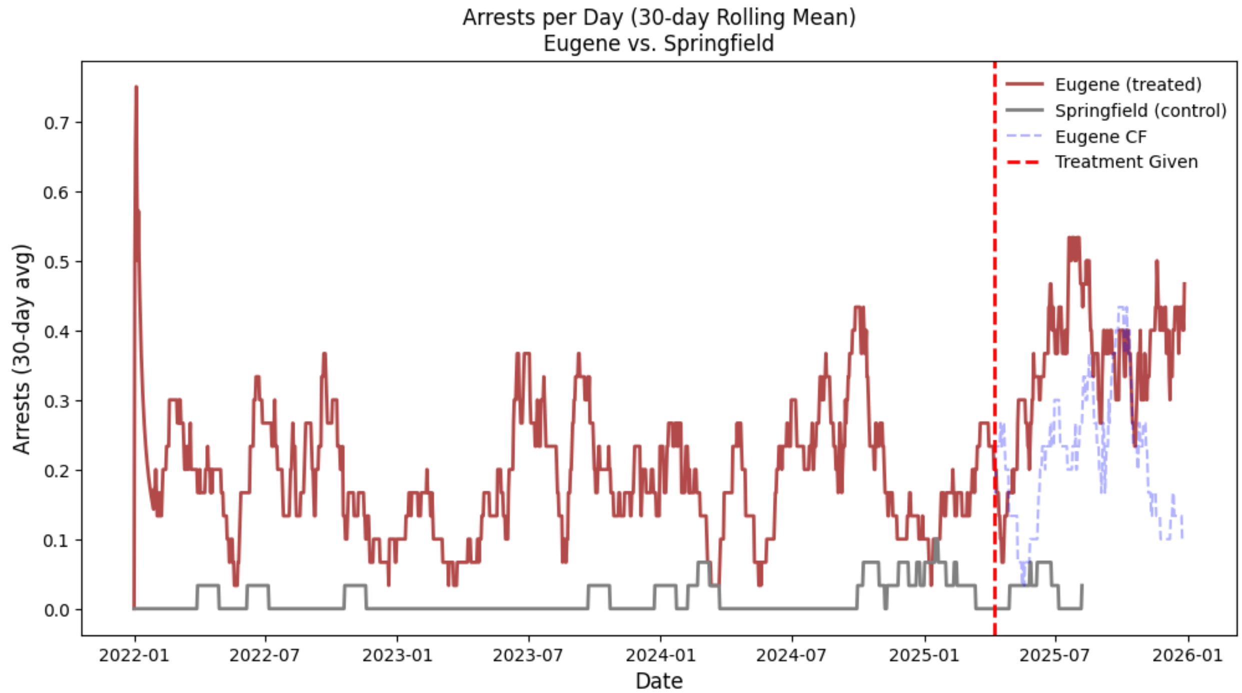

Figure 9: DiD Lineplot#2: Arrest Outcome

From the t-test conducted, the null hypothesis (as said before) was that the monthly average volume of arrest outcomes are relatively the same. After conducting a t-test with a p-value ¡ 0.001, we can reject the null hypothesis and accept the alternative hypothesis that the post-defunding monthly average of arrest outcomes is larger, therefore saying that the increase in arrests is statistically significant.

4.3 Summary of Results

From these results, in terms of the volume of welfare check calls being called in, Eugene is seeing less than what it is supposed to be at. After doing the t-test, the result was that this drop was statistically significant. In terms of the arrest outcomes, Eugene is seeing more than what it is supposed to be at. The t-test says that this rise in arrest outcomes is also statistically significant.

5 Discussion

What can be done to get these numbers to where they are supposed to be? What can the city do to decrease the number of arrests concluding welfare check calls and increase the volume of welfare checks?

5.1 Recommendations

One recommendation is to bring law enforcement (EPD) to the standards of CAHOOTS. Before defunding, CAHOOTS had many options to assist those in need: the ability to transport people, hand off people to emergency services, and help the person on scene, as every CAHOOTS van had a social worker and a registered nurse. Now that CAHOOTS is no longer in operation, law enforcement is the only unit able to assist those on the streets. Some officers may not have the same training as CAHOOTS workers, which means treatment may not be given as easily. If officers were given similar training, fewer arrest outcomes would likely follow.

In terms of call volume, the city could reinstate CAHOOTS. Before defunding, people in Eugene trusted that if a welfare check was called in, someone would arrive to help. Now that CAHOOTS is no longer in operation, that trust is gone and there is confusion about who to call. If CAHOOTS were reinstated, that trust would slowly be rebuilt and more people would be inclined to call in welfare checks, solving the issue of declining call volume.

5.2 Limitations

All research has limitations and this is no different. The first limitation is the lack of data after CAHOOTS was defunded. Before, there were 10 years of data (from 2015 to 2025) and after, only 8 months. More data is needed to make stronger comparisons and conclusions. A second limitation is that the recommendations above are based on theoretical thinking. Any changes made to the city along these lines would need input from an expert or someone with extensive field experience.